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Clean numerical simulation: a new strategy to obtain reliable solutions of chaotic dynamic systems*

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Abstract It is well-known that chaotic dynamic systems, e.g., three-body system and turbulent flow, have sensitive dependence on the initial conditions (SDIC). Unfortunately, numerical noises, i.e., truncation error and round-off error, always exist in practice. Thus, due to the SDIC, the long-term accurate prediction of chaotic dynamic systems is practically impossible. In this paper, a new strategy for chaotic dynamic systems, i.e., the clean numerical simulation (CNS), is briefly described, and applied to a few Hamiltonian chaotic systems. With negligible numerical noises, the CNS can provide convergent (reliable) chaotic trajectories in a long enough interval of time. This is very important for Hamiltonian systems, and thus should have many applications in various fields. It is found that the traditional numerical methods in double precision cannot give not only reliable trajectories but also reliable Fourier power spectra and autocorrelation functions (ACFs). In addition, even the statistic properties of chaotic systems cannot be correctly obtained by means of traditional numerical algorithms in double precision, as long as these statistics are time-dependent. The CNS results strongly suggest that one had better be very careful on the direct numerical simulation (DNS) results of statistically unsteady turbulent flows, although DNS results often agree well with experimental data when the turbulent flow is in a statistical stationary state.

Key words chaos, numerical noise, clean numerical simulation (CNS), reliability of computation

Chinese Library Classification O415.6
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1 Introduction

Poincaré[1] discovered in mathematics that some dynamic systems have sensitivity dependence on the initial conditions (SDIC), i.e., a tiny difference in the initial conditions might lead to a significant variation in the solution after a long enough time. The SDIC was rediscovered by Lorenz[2] in 1960s by means of digit computers. He made the SDIC more popular by giving it a famous name "butterfly-effect", i.e., a hurricane in North America might be created by the flapping of the wings of a distant butterfly in South America several weeks earlier. To make matters even worse, Lorenz[3–4] further found that such chaotic simulations are sensitive not only to the initial conditions but also to the numerical algorithms, i.e., different numerical schemes with different time steps might lead to completely different numerical results of chaos. Lorenz's conclusions were confirmed by many researchers[5–7]. This is easy to understand since numerical noises, i.e., truncation error and round-off error, inherently exist at each time step for all numerical schemes, which will be enlarged exponentially due to the SDIC of chaos[2]. These numerical phenomena lead to the intense arguments[8–9] about the reliability of numerical simulations of chaotic systems. A few researchers even believed that "all chaotic responses are simply numerical noises and have nothing to do with the solutions of differential equations"[8]. Thus, due to the SDIC (or the butterfly-effect), it is indeed a challenge to accurately simulate the chaotic solution of a nonlinear dynamic system in a long interval of time[10–11].

It is widely believed that turbulence[12–15] is chaotic. Therefore, statistics is commonly used in the study of turbulence, and the direct numerical simulation (DNS)[16], which solves the Navier-Stokes equations without averaging or approximation but with all essential scales of motion, has played an important role in turbulence statistics. However, due to the SDIC, tiny numerical noises, which grow exponentially in time, may lead to spurious results. Yee et al.[17] reported in 1999 that the DNS could produce a spurious solution, which was completely different from the physical solution of their considered case. Wang and Rosa[18] demonstrated a spurious evolution of turbulence originated from the round-off error in the DNS. For more examples of spurious numerical simulations, please refer to Refs. [19] and [20]. Moreover, the analyses of chaotic dynamics were numerically studied in the non-linear vibrations of spatial structures such as shells[21–22], plates[23], and beams[24–26]. The qualitative theory of differential equations has been widely used in the study of chaotic dynamic systems. Chaotic dynamic systems have been investigated by means of the Poincaré maps, Lyanponuv exponent, phase portraits, Fourier and wavelet power spectra, autocorrelation functions (ACFs), etc.[23–31]. Therefore, it is necessary to verify the reliability of these results for chaotic dynamic systems.

Recently, a new numerical strategy, i.e., "clean numerical simulation" (CNS)[32–42], was proposed to gain the reliable simulations of chaotic dynamic systems in a finite but long enough interval of time. The CNS is based on an arbitrary Taylor series method (TSM)[43–47] and the multiple precision[48] in arbitrary accuracy, plus a kind of convergence verification by means of additional simulations with even less numerical noises. Note that the CNS is a kind of remixing of some known methods, each of which was used separately for other purposes. By means of the CNS, numerical noises can be so greatly reduced that they are much smaller than the "true" solution where the numerical noises are negligible in a given interval of time even for chaotic dynamic systems. For example, using the traditional Runge-Kutta method in double precision, one can gain the convergent chaotic results of the Lorenz equation only in a few dozens of time intervals; however, using the CNS, Liao and Wang[35] successfully obtained a convergent reliable chaotic solution to the Lorenz equation in an interval $[0, 10\,000]$, which is more than 300 times larger than that given by the traditional Runge-Kutta method in double precision. Recently, Li and Liao[40] demonstrated that more than 243 periodic three-body orbits were found by means of the CNS than by means of the conventional Runge-Kutta method in double precision. A generalized Kepler's third law for three-body problem was found by Li and Liao[40] and extended to a generalized Kepler's third law for n-body problem by Sun[49] with

a dimensional analysis. This illustrates the validity of the CNS for the reliable simulations of chaotic dynamic systems with the SDIC.

In this paper, we further apply the CNS to some chaotic Hamiltonian systems, such as the Hénon-Heiles system and the famous three-body problem. The effects of numerical noises on the statistic results of chaotic dynamic systems are investigated. The strategy of the CNS is briefly described in Section 2. The applications in chaotic Hamiltonian systems are given in Section 3. The effects of numerical noises on the chaotic dynamic systems are investigated in Section 4. The concluding remarks are provided in Section 5.

2 A brief description of the CNS

Numerical noises, i.e., truncation and round-off errors, always exist in computer-generated numerical simulations, forever. In order to gain the reliable simulations of chaotic dynamic systems in an arbitrarily long (but finite) interval of time, we must reduce the numerical noises to a desired tiny level.

Consider a nonlinear dynamic system

$$\frac{\mathrm{d}\boldsymbol{y}}{\mathrm{d}t} = \boldsymbol{F}(t, \boldsymbol{y}), \quad \boldsymbol{y}(0) = \boldsymbol{y}_0, \quad t \in \mathbb{R}, \quad \boldsymbol{y}_0,\, \boldsymbol{y},\, \boldsymbol{F} \in \mathbb{R}^n, \tag{1}$$

where t is the time, $\boldsymbol{y}(t)$ is the vector of unknown functions with $\boldsymbol{y}_0$ being its initial value, $\boldsymbol{F}$ denotes the vector of the nonlinear functions. Write $t_{n+1} = t_n + h$, where h is the time step. Then, we have the Mth-order Taylor expansion as follows:

$$\boldsymbol{y}(t_{n+1}) \simeq \boldsymbol{y}(t_n) + \frac{\mathrm{d}\boldsymbol{y}(t_n)}{\mathrm{d}t}h + \frac{1}{2!}\frac{\mathrm{d}^2\boldsymbol{y}(t_n)}{\mathrm{d}t^2}h^2 + \cdots + \frac{1}{M!}\frac{\mathrm{d}^M\boldsymbol{y}(t_n)}{\mathrm{d}t^M}h^M, \tag{2}$$

where the Taylor coefficients can be calculated in a recursive way by using Eq. (1).

Obviously, the higher the order M of the Taylor expansion in Eq. (2) is, the smaller the truncation error is, as long as the time step h is within the radius of convergence. Besides, the round-off error can be reduced to any required level by means of the data in multiple precision[48] with a large enough number of digits. Therefore, with enough resources of computation, one can reduce the numerical noises to an arbitrarily tiny level! In addition, the convergence (reliability) of a computer-generated result is verified by comparing it with an additional simulation with even smaller numerical noises, which is a well-known approach widely used in the field of uncertainty quantification. Therefore, the CNS is based on the remixing of some known methods/technologies. However, it is interesting that such a kind of remixing of the old methods could bring us something new/different, as mentioned below.

3 Applications of the CNS in some chaotic Hamiltonian systems

There are many chaotic Hamiltonian systems in physics, from the microscopic quantum chaos[50] to the macroscopic solar system[51]. Nowadays, symplectic integrators (SIs) have been widely used to numerically solve the Hamiltonian systems[52–54] such as spin systems[52] and the solar system[55]. Because an SI possesses a Hamiltonian as a conserved quantity often corresponding to the total energy of the system, it has been widely used to calculate the long-term evolution of chaotic Hamiltonian systems with relatively large time steps.

Are long-term orbits of chaotic Hamiltonian systems obtained by SIs indeed reliable? To check this, one should be able to gain the convergent chaotic solution in a long enough interval. Here, we use the CNS to gain the reliable orbits of the Hénon-Heiles system and the famous three-body problem, and compare the obtained results with those given by the SIs in double precision.

3.1 SI

Consider a generic Hamiltonian system

$$\dot{\boldsymbol{p}} = -\frac{\partial H}{\partial \boldsymbol{q}}, \quad \dot{\boldsymbol{q}} = \frac{\partial H}{\partial \boldsymbol{p}}, \tag{3}$$

where $\boldsymbol{q}$ denotes the position coordinate vector, $\boldsymbol{p}$ is the momentum coordinate vector, $H(\boldsymbol{p}, \boldsymbol{q})$ is the Hamiltonian corresponding to the total energy of the system. Assume that the Hamiltonian is separable, i.e., $H(\boldsymbol{p}, \boldsymbol{q}) = T(\boldsymbol{p}) + V(\boldsymbol{q})$, where $T(\boldsymbol{p})$ is the kinetic energy, and $V(\boldsymbol{q})$ denotes the potential energy. We use here the classical 4th-order explicit SI as follows:

$$\boldsymbol{q}_i = \boldsymbol{q}_{i-1} + hc_i \frac{\partial T(\boldsymbol{p}_{i-1})}{\partial \boldsymbol{p}}, \tag{4}$$

$$\boldsymbol{p}_i = \boldsymbol{p}_{i-1} - hd_i \frac{\partial V(\boldsymbol{q}_i)}{\partial \boldsymbol{q}}, \tag{5}$$

where $i = 1, 2, 3, 4$, h denotes the time step, and

$$\begin{cases} c_1 = c_4 = \dfrac{1}{2(2-2^{1/3})}, \quad c_2 = c_3 = \dfrac{1-2^{1/3}}{2(2-2^{1/3})}, \\ d_1 = d_3 = \dfrac{1}{2-2^{1/3}}, \quad d_2 = -\dfrac{2^{1/3}}{2-2^{1/3}}, \quad d_4 = 0 \end{cases} \tag{6}$$

are constant coefficients. For details, please refer to Refs. [56] and [57].

3.2 Hénon-Heiles system

The motion of stars orbiting in a plane about the galactic center is governed by the so-called Hénon-Heiles Hamiltonian system of equations[58] as follows:

$$\ddot{x}(t) = -x(t) - 2x(t)y(t), \quad \ddot{y}(t) = -y(t) - x^2(t) + y^2(t). \tag{7}$$

Here, the Hamiltonian is the total energy, i.e., $H = T(\dot{x}, \dot{y}) + V(x, y)$, where $T(\dot{x}, \dot{y}) = \frac{1}{2}\big(\dot{x}^2 + \dot{y}^2\big)$ is the kinetic energy, and

$$V(x, y) = \frac{1}{2}\Big(x^2 + y^2 + 2x^2 y - \frac{2}{3}y^3\Big)$$

is the potential energy. As pointed out by Sprott[59], the solution is chaotic for some initial conditions, e.g.,

$$x(0) = \frac{14}{25}, \quad y(0) = 0, \quad \dot{x}(0) = 0, \quad \dot{y}(0) = 0, \tag{8}$$

which are considered in this paper.

First of all, we simulate the chaotic orbits of the Hénon-Heiles system (see Eq. (7)) with the initial condition (see Eq. (8)) in the interval $[0, 1\,000]$ by means of the 4th-order SI in double precision and the time steps $h = 0.001$, 0.000 1, and 0.000 01, respectively. As shown in Fig. 1, the relative energy errors of all these simulations given by the 4th-order SI are rather small in the whole interval $[0, 1\,000]$, i.e., less than 10^{-12}. However, as shown in Fig. 2, their orbits depart from each other quickly. The orbits given by the SI with the time steps $h = 0.001\ 00$ and $h = 0.000\ 100$ separate at about $t = 280$, and the orbits with the time steps $h = 0.000\ 10$ and $h = 0.000\ 01$ separate at about $t = 310$. Therefore, none of these trajectories given by the SI are reliable in the interval $[0, 1\,000]$, though the Hamiltonian, i.e., the total energy of the system, is conserved quite well.

Liao[60] successfully used the CNS to gain the convergent trajectories of the chaotic Hénon-Heiles system (see Eq. (7)) with the initial condition (see Eq. (8)) in the interval $[0, 2\,000]$ by

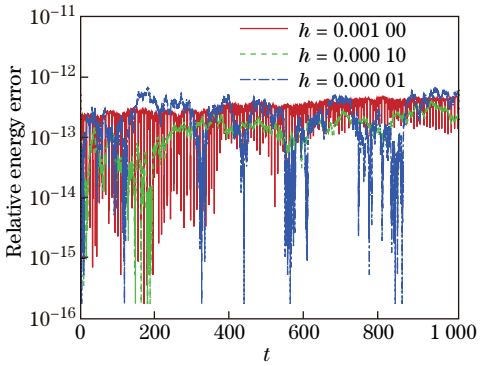


Fig. 1 Evolution of the relative energy error for the Hénon-Heiles system (see Eqs. (7)–(8)) given by the 4th-order SI in double precision and with different h (color online)

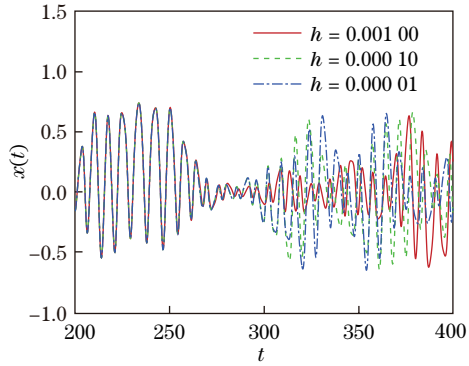


Fig. 2 Simulations of $x(t)$ of the Hénon-Heiles system (see Eqs. (7)–(8)) given by the 4th-order SI in double precision with different h (color online)

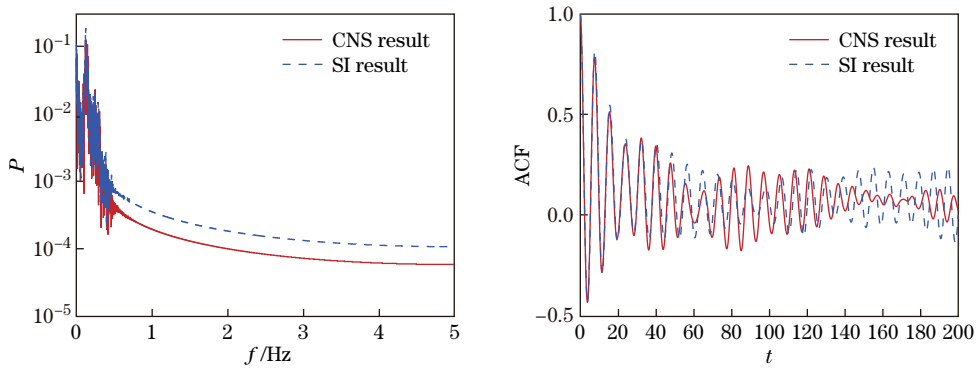
means of the CNS with the 70th-order Taylor expansion, where the data are in the 140-digit multiple precision, and the time step is $h = 0.1$. Following Liao[60], we gain a reliable convergent orbit for Eqs. (7) and (8) in the interval $[0, 1\,000]$ by means of the CNS with the 50th-order Taylor expansion, where the data are in the 100-digit multiple precision, and the time step is $h = 0.01$, as listed in Table 1. Note that the orbits given by the CNS at the 30th-order ($M = 30$) of Taylor expansion are convergent in the accuracy of 17 significance digits, and the corresponding deviation from the total energy is very small, i.e., 1.8×10^{-73}, which is 50 orders of magnitude less than that given by the SI (see Fig. 1). Therefore, unlike the SI in double precision, the CNS can give convergent reliable orbits for the chaotic Hénon-Heiles system with very small deviation from the total energy, i.e., energy preserving, in a long interval $[0, 1\,000]$. This illustrates that the CNS can give more reliable orbits for chaotic Hamiltonian systems than the SI.

Table 1 x and deviation from the total energy at $t = 1\ 000$ by means of the CNS with different orders of Taylor's expansion, where the data are in the 100-digit multiple precision, and the time step is $h = 0.001$

Order	$x(t)$	Deviation from the total energy H
20	−0.049 000 000 000 000 00	9.1×10^{-48}
25	−0.049 154 038 397 848 00	2.0×10^{-60}
30	−0.049 154 038 397 848 44	1.8×10^{-73}
40	−0.049 154 038 397 848 44	2.6×10^{-96}

As shown in Fig. 3(a), the Fourier power spectrum of $x(t)$ ($t \in [0, 1\,000]$) of the Hénon-Heiles system given by CNS results is different from that obtained by the 4th-order SI in double precision and with the time step $h = 0.000\ 01$ when the frequency f is higher than 0.4 Hz. This implies that the Fourier power spectra given by the SI in double precision are unreliable when $f > 0.4$ Hz. The ACF of $x(t)$ ($t \in [0, 1\,000]$) of the Hénon-Heiles system obtained by the 4th-order SI has a big difference from the CNS result when $t > 50$, as shown in Fig. 3(b). It suggests that the 4th-order SI in double precision cannot give reliable Fourier power spectra and ACFs.

Why? This is mainly due to the butterfly-effect of chaos, i.e., the tiny difference in the initial condition enlarges exponentially[2]. Here, we should mention that Liao[60] used the CNS



(a) Fourier power spectra (b) ACF

Fig. 3 Fourier power spectra and ACF of $x(t)$ ($t \in [0, 1\,000]$) of the Hénon-Heiles system given by the CNS results and the 4th-order SI in double precision and with the time step $h = 0.000\,01$ (color online)

to gain a convergent solution of the chaotic Hénon-Heiles system (see Eq. (7)) with the initial condition

$$x(0) = \frac{14}{25}, \quad y(0) = 10^{-60}, \quad \dot{x}(0) = 0, \quad \dot{y}(0) = 0, \tag{9}$$

which has a tiny difference from Eq. (8), and found that this tiny difference in the initial condition indeed led to the completely different orbits. Unfortunately, the numerical noises, i.e., truncation error and round-off error, always exist for the numerical schemes including the SIs in double precision, where, in general, the round-off error is at the level 10^{-16}, which is much greater than 10^{-60}. Therefore, it is reasonable that, even SIs cannot give convergent reliable long-term trajectories, Fourier power spectra, and ACFs of the chaotic Hénon-Heiles system in some cases.

3.3 Three-body problem

Now, let us consider another Hamiltonian system, i.e., the famous three-body problem, governed by the Newtonian gravitational law and the motion equations as follows:

$$\ddot{x}_{k,i} = \sum_{j=1, j\neq i}^{3} \frac{G m_j (x_{k,j} - x_{k,i})}{R_{i,j}^3}, \quad k = 1, 2, 3, \tag{10}$$

where m_j is the mass of the jth body, and

$$R_{i,j} = \left(\sum_{k=1}^{3} (x_{k,j} - x_{k,i})^2 \right)^{\frac{1}{2}}.$$

$\boldsymbol{r}_i = (x_{1,i}, x_{2,i}, x_{3,i})$ denotes the position of the ith body. Without loss of generality, let us consider here the case in which $m_1 = m_2 = m_3 = 1$ and $G = 1$ and the initial condition

$$\begin{cases} \boldsymbol{r}_1 = (1/10, 0, -1), \quad \boldsymbol{r}_2 = (0, 0, 0), \quad \boldsymbol{r}_3 = (0, 0, 1), \\ \dot{\boldsymbol{r}}_1 = (0, -1, 0), \quad \dot{\boldsymbol{r}}_2 = (1, 1, 0), \quad \dot{\boldsymbol{r}}_3 = (-1, 0, 0). \end{cases} \tag{11}$$

The 4th-order SI in double precision is used to gain the chaotic orbits of the three-body system (see Eqs. (10)–(11)) in the interval $[0, 1\,000]$ by means of the time steps $h = 10^{-4}$, 10^{-5}, and 10^{-6}. Since the three-body problem is a Hamiltonian system, its total energy must be conserved for a reliable simulation. As shown in Fig. 4, the deviation from the total energy of

the three-body problem given by the SI in double precision is indeed rather small in the whole interval $[0, 1\,000]$, at a level less than 10^{-8}. Unfortunately, this cannot guarantee the reliability of the chaotic trajectory of the three-body system. A shown in Fig. 5, the x position of Body-1 given by the time step $h = 10^{-4}$ departs at $t \approx 270$ from that given by $h = 10^{-5}$, and the x positions given by $h = 10^{-5}$ and $h = 10^{-6}$ depart from each other at $t \approx 310$. Therefore, the long-term evolution of the chaotic orbits of the three-body problem (see Eqs. (10)–(11)) given by the 4th-order SI in double precision is not reliable in the interval $[0, 1\,000]$.

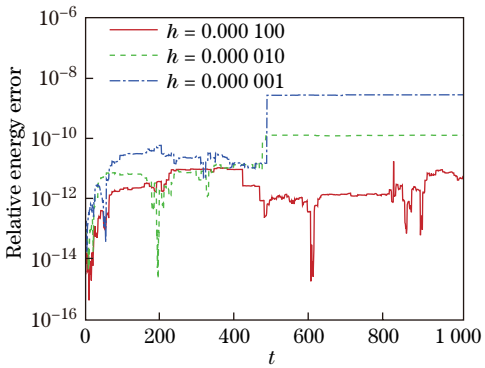


Fig. 4 Evolution of the deviation from the total energy of the three-body problem given by the 4th-order SI in double precision with different time steps h (color online)

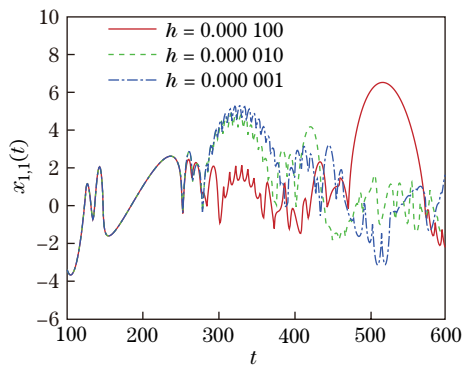


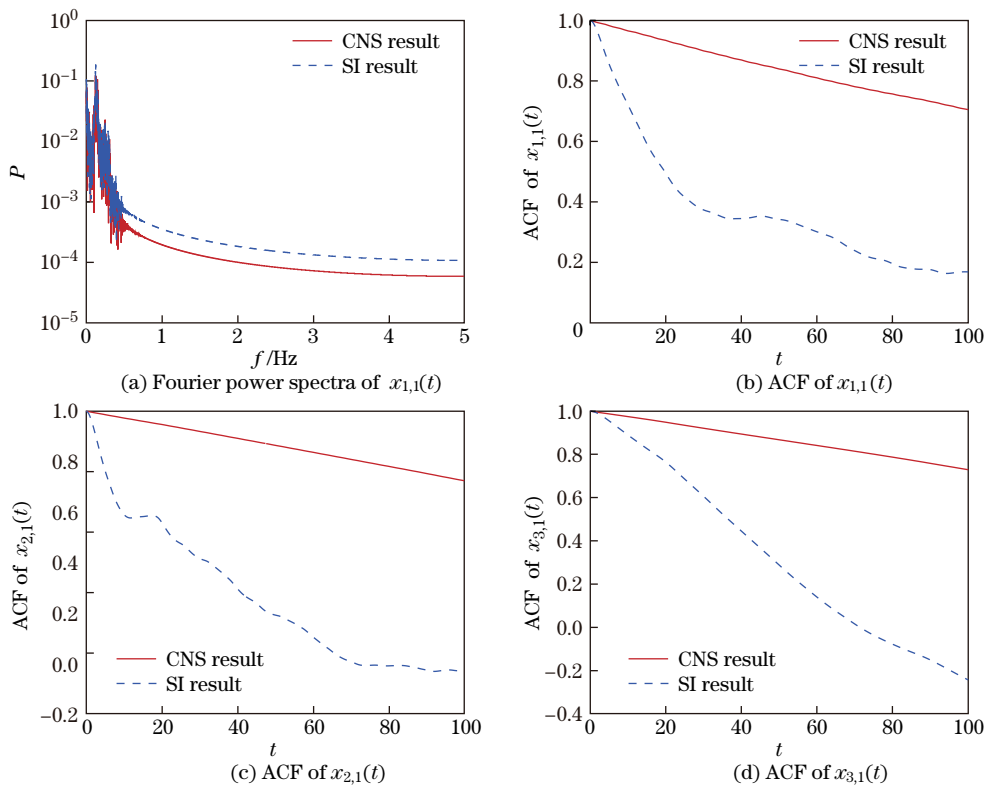
Fig. 5 x position of Body-1 given by the 4th-order SI in double precision with different time steps (color online)

By means of the CNS with the 80th-order Taylor expansion, where the data were in 300-digit multiple precision and the time step was $h = 10^{-2}$, Liao[34] successfully gained the reliable orbits of a similar chaotic three-body problem in the interval $[0, 1\,000]$. Similarly, following Liao[34], we obtain a convergent long-term evolution of the three-body system (see Eqs. (10)–(11)) in the interval $[0, 1\,000]$ by means of the CNS. Indeed, by means of the CNS with the up-to 80th-order Taylor expansion and the data in the 300-digit multiple precision with the time step $h = 10^{-3}$, we gain the convergent orbits of the three-body system in the time interval $[0, 1\,000]$, as shown in Table 2. Note that the orbits at $t = 1\,000$ given by the CNS at the 40th- and 60th-orders of Taylor expansion are convergent in the accuracy of 37 and 12 significance digits, respectively. Especially, the orbits given by the CNS at the 70th- and 80th-orders of Taylor expansion are convergent in the accuracy of 16 significance digits, with very small deviations from the total energy, i.e., at the levels 10^{-31} and 10^{-35}, respectively. Thus, unlike the SI, the CNS can give a reliable long-term evolution for the chaotic orbits of the three-body system, together with a rather small deviation from the total energy. It should be emphasized that, for the three-body problem, it is very important to give an accurate prediction of orbits. As shown in Fig. 6(a), the Fourier power spectrum of $x_{1,1}(t)$ ($t \in [0, 1\,000]$) of Body-1 of the three-body system given by the CNS has a large difference from the power spectrum obtained by the 4th-order SI in double precision and with the time step $h = 0.000\,001$ when $f > 0.4$ Hz. This implies that the Fourier power spectra given by the SI in double precision are unreliable when $f > 0.4$ Hz. Besides, the ACFs of $x_{1,1}(t)$, $x_{2,1}(t)$, and $x_{3,1}(t)$ ($t \in [0, 1\,000]$) of Body-1 of the three-body system obtained by the 4th-order SI are totally different from those obtained by the CNS, as shown in Figs. 6(b), 6(c), and 6(d). It also suggests that the 4th-order SI in double precision cannot give reliable Fourier power spectra and ACFs.

These two examples illustrate that, for some chaotic Hamiltonian systems, even the SIs in double precision cannot give convergent (reliable) numerical simulations of trajectories, Fourier

Table 2 x positions of Body-1 of the three-body system (see Eqs. (10)–(11)) at $t = 1\ 000$ given by the CNS with different orders of Taylor expansion, where the data are in 300-digit multiple precision, and the time step is $h = 10^{-3}$

Order	$x_{1,1}(t)$	Deviation from the total energy
40	−16.800 000 000 000 00	1.63×10^{-17}
50	−16.828 690 000 000 00	7.88×10^{-22}
60	−16.828 692 538 900 00	2.05×10^{-26}
70	−16.828 692 538 941 94	1.12×10^{-31}
80	−16.828 692 538 941 94	1.90×10^{-35}



(a) Fourier power spectra of $x_{1,1}(t)$ (b) ACF of $x_{1,1}(t)$ (c) ACF of $x_{2,1}(t)$ (d) ACF of $x_{3,1}(t)$

Fig. 6 Fourier power spectra of $x_{1,1}(t)$ and ACFs of $x_{1,1}(t)$, $x_{2,1}(t)$, and $x_{3,1}(t)$ of Body-1 of the three-body system given by the CNS and the 4th-order SI in double precision and with the time step $h = 0.000\ 001$ (color online)

power spectra, and ACFs in a long enough interval of time. In fact, to the best of our knowledge, all traditional numerical algorithms in double precision cannot give convergent (reliable) long-term prediction of trajectories for chaotic dynamic systems. But, the CNS can do it. So, although the CNS is a remixing of some well-known methods/technologies, it can indeed bring us something completely new/different!

4 Effects of numerical noises on statistics of chaotic systems

It is widely believed that, although traditional algorithms in double precision cannot give correct trajectories of chaotic systems, they, however, might correctly give statistical properties.

Is this indeed true? In this section, we will illustrate the effects of numerical noises on the statistic of chaotic dynamic systems via an example.

It is well-known that the famous Lorenz equation[2] is a quite simplified model of the Rayleigh-Bénard (RB) flow of a viscous fluid. From the exact Navier-Stokes equations for the two-dimensional RB flow

$$\nabla^2\frac{\partial\psi}{\partial t}+\frac{\partial(\psi,\nabla^2\psi)}{\partial(x,z)}-\sigma\frac{\partial\theta}{\partial x}-\sigma\nabla^4\psi=0, \tag{12}$$

$$\frac{\partial\theta}{\partial t}+\frac{\partial(\psi,\theta)}{\partial(x,z)}-Re\frac{\partial\psi}{\partial x}-\nabla^2\theta=0, \tag{13}$$

where ψ denotes the stream function, θ is the temperature departure from a linear variation background, t is the time, x is the horizontal coordinate, z is the vertical coordinate, σ is the Prandtl number, and Re is the Rayleigh number. Saltzman[61] deduced a family of highly truncated dynamic systems of different degrees of freedom (DOFs), where the famous Lorenz equation[2] was the simplest one among them. In the case of the Prandtl number $\sigma=10$, the highly truncated dynamic system of three DOFs reads

$$\begin{cases}\dot{A}=-148.046A-1.500D, \quad \dot{G}=27.916AD-39.479G,\\ \dot{D}=-13.958AG-1\ 460.631\lambda A-14.805D,\end{cases} \tag{14}$$

where $\lambda=Re/Re_{\rm c}$ is the dimensionless Rayleigh number, $Re_{\rm c}$ is the critical Rayleigh number, A and D represent the cellular streamline and thermal fields for the Rayleigh critical mode, respectively, and G denotes the departure of the vertical variation. For details, please refer to Ref. [61].

Similarly, Saltzman[61] gave the highly truncated dynamic system of five DOFs as

$$\begin{cases}\dot{A}=23.521BC-1.500D-148.046A, \quad \dot{B}=-22.030AC-186.429B,\\ \dot{C}=1.561AB-400.276C, \quad \dot{G}=27.916AD-39.479G,\\ \dot{D}=-13.958AG-1\,460.631\lambda A-14.805D,\end{cases} \tag{15}$$

and the highly truncated dynamic system of seven DOFs as

$$\begin{cases}\dot{A}=23.521BC-1.500D-148.046A, \quad \dot{B}=-22.030AC-1.589E-186.429B,\\ \dot{C}=1.561AB-0.185F-400.276C, \quad \dot{G}=27.916AD+37.220BE-39.479G,\\ \dot{D}=-16.284CE-16.284BF-13.958AG-1\ 460.631\lambda A-14.805D,\\ \dot{E}=16.284CD-16.284AF-18.610BG-1\ 947.508\lambda B-18.643E,\\ \dot{F}=16.284AE+16.284BD-486.877\lambda C-40.028F.\end{cases} \tag{16}$$

All of them are deterministic equations with chaotic solutions, and are simplified models for the two-dimensional RB flow. It should be emphasized here that, for any given initial condition, we can gain reliable convergent numerical results for the chaotic solutions of these models in a finite but long enough interval of time by means of the CNS.

In physics, the two-dimensional RB flow with a large enough Rayleigh number Re is an evolutionary process from an initial equilibrium state to turbulence after a long enough time, mainly because the flow is unstable and besides micro-level physical uncertainty, e.g., thermal fluctuation, always exists. Such a kind of initial micro-level physical uncertainties due to thermal fluctuation can be expressed by Gaussian random data[13]. Mathematically, due to the SDIC,

the chaotic solutions of these simplified models should be dependent on the random initial conditions. This is true in physics, since all experimental measurements of RB turbulent flows are different. However, due to the butterfly-effect, one cannot obtain convergent (reliable) trajectories of these chaotic systems in a long enough interval of time by means of the traditional numerical algorithms in double precision. Therefore, the above-mentioned equations provide us a few simplified models to investigate the effects of numerical errors on the statistics of such a kind of chaotic dynamic systems.

Without loss of generality, let us first consider the deterministic three DOFs (see Eq. (14)) with the random initial conditions (physically related to the micro-level thermal fluctuation) in a normal distribution in case of $\lambda = 28$, corresponding to a turbulent flow. Considering the thermal fluctuation, we study here such a kind of random initial conditions in normal distribution with the mean

$$\langle A(0)\rangle = 1, \quad \langle D(0)\rangle = 10^{-3}, \quad \langle G(0)\rangle = 10^{-3}$$

and the standard deviation

$$\sigma_0 = \sqrt{\langle A^2(0)\rangle} = \sqrt{\langle D^2(0)\rangle} = \sqrt{\langle G^2(0)\rangle} = 10^{-30}.$$

Here, the standard deviation is related to the micro-level thermal fluctuation, which is much smaller than the numerical noises of traditional numerical algorithms in double precision. However, by means of the CNS with the numerical noises much smaller than thermal fluctuation, we can gain convergent (reliable) chaotic propagations (trajectories) of micro-level physical uncertainties of these systems for any given initial condition. Let

$$\langle A(t)\rangle = \frac{1}{N}\sum_{i=1}^{N} A_i(t), \tag{17}$$

$$\sigma_A(t) = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(A_i(t) - \langle A(t)\rangle)^2} \tag{18}$$

denote the sample mean and unbiased estimate of the standard deviation of $A(t)$ of these reliable simulations of trajectories, respectively, where N is the number of samples.

We obtain 2 000 samples of reliable numerical simulations of the system of three DOFs (see Eq. (14)) in the time interval $[0, 10]$ by means of the CNS with the 80th-order Taylor series ($M = 80$), the 90 decimal-digit precision ($K = 90$) for every datum, and the time step $h = 0.001$. It is found that the numerical errors can be decreased to be much smaller than the micro-level physical uncertainties in the time interval $[0, 10]$ under consideration. These numerical simulations are so accurate that we can consider them as the "true" solutions of the chaotic dynamic system (see Eq. (14)), which can be used to investigate the effects of numerical noises on statistic computations of chaotic systems. In this way, we successfully distinguish/separate the "true" (convergent) chaotic solutions having physical meanings from the numerical noises having no physical meanings! Note that, for the chaotic dynamic system (see Eq. (14)), the mean and standard deviation of $A(t)$ with 1 000 samples are almost the same as those with 2 000 samples, as shown in Fig. 7. Thus, it is enough for us to use 2 000 samples in this paper.

Obviously, the larger the order M of the Taylor series in the frame of the CNS is, the smaller the truncation error is. For example, the traditional 4th-order Runge-Kutta's method corresponds to the CNS with the 4th-order Taylor series expansion. Thus, to investigate the effects of the truncation error, we use here the 10th-order Taylor series, i.e., $M = 10$, but remain the 90 decimal-digit multiple precision for every datum so as to make the round-off

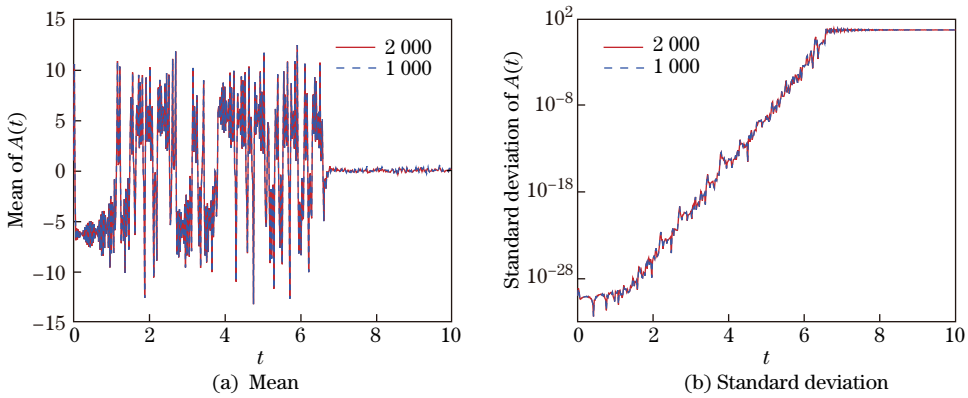


Fig. 7 Mean and standard deviation of $A(t)$ of the chaotic system of three DOFs (see Eq. (14)) with different numbers of samples gained by means of the CNS (color online)

error negligible. In this way, the round-off error is negligible in the considered interval of time $t \in [0, 10]$ so that the effect of the truncation error can be investigated independently. Note that our reliable CNS results are gained by means of $M = 80$, i.e., the 80th-order Taylor series, whose truncation errors are negligible in the considered interval of time $t \in [0, 10]$. However, when $M = 10$, the truncation error cannot be negligible in $[0, 10]$.

The reliable mean (the solid red line, given by $M = 80$) and the unreliable result (the bashed blue line, given by $M = 10$) of $A(t)$ of the chaotic dynamic system of three DOFs (see Eq. (14)) are shown in Fig. 8. From the figure, we can see that the reliable mean of $A(t)$ given by the CNS becomes stable when $t > 7.0$, but is time-dependent when $t \leqslant 7.0$. However, as shown in Fig. 8, the unreliable mean of $A(t)$ given by $M = 10$ (with the considerable truncation error) has a noticeable difference from the reliable mean given by the CNS within $2.5 < t < 7.0$. It suggests that the truncation error has a great effect on the unsteady statistical quantities when the chaotic dynamic system is in a transition stage from an equilibrium state to another one. Note that the mean of $A(t)$ given by $M = 10$ agrees well with the reliable CNS result when $t > 7.0$. Thus, the truncation error seems to have no effect on the time-independent statistics

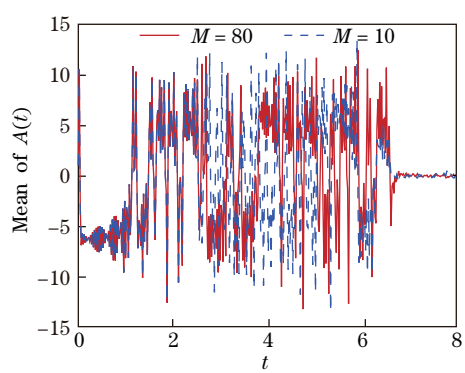


Fig. 8 Effects of the truncation error on the mean of $A(t)$ of the chaotic dynamic system of three DOFs (see Eq. (14)) using the time step $h = 10^{-3}$ and the 90 decimal-digit precision for every datum, i.e., with the negligible round-off error, where the solid line in red means the reliable mean of $A(t)$ given by the CNS using $M = 80$, i.e., with the negligible truncation error, while the dashed line in blue denotes the mean of $A(t)$ given by $M = 10$, i.e., with the considerable truncation error (color online)

of chaotic dynamic systems in an equilibrium state. The scatter diagrams of the probability distribution function (PDF) in the AD-plane of the chaotic dynamic system of three DOFs (see Eq. (14)) given by the two numerical schemes, i.e., $M = 10$ and $M = 80$, are as shown in Fig. 9, where the PDF is obtained by using the Gaussian kernel density estimator. Note that the PDFs given by the two numerical approaches are almost the same at $t = 8.00$. However, at $t = 6.60$ and $t = 6.80$, the PDFs given by $M = 10$ are quite different from those given by the CNS with $M = 80$. This supports our previous conclusion that the truncation error has a significant effect on the unsteady statistics of the chaotic dynamic system but no effect on steady ones. This might be the reason why many DNS results for fully developed turbulence agree well with experimental data.

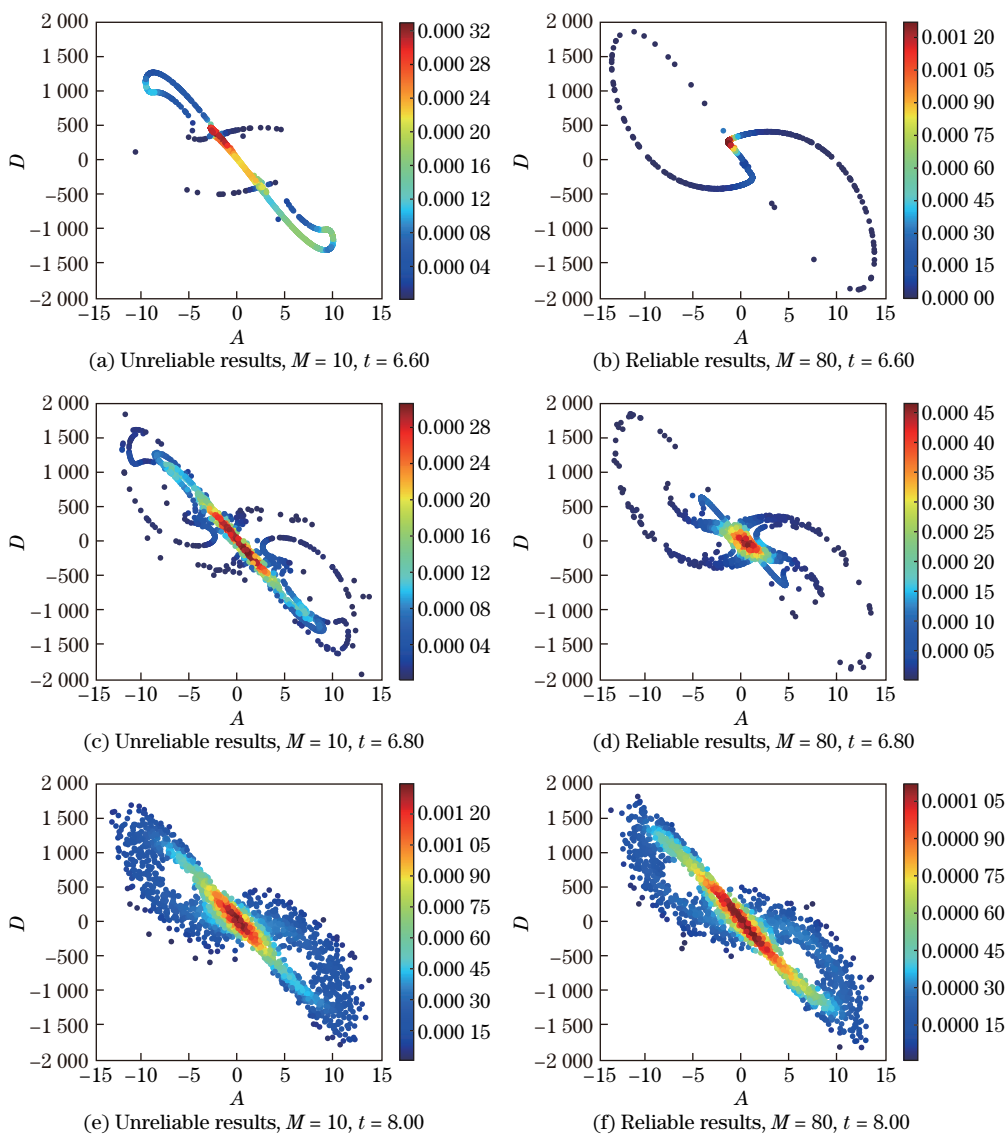


Fig. 9 Scatter diagrams of the PDF in the AD-plane of the system of three DOFs (see Eq. (14)) at different time by means of the time step $h = 10^{-3}$, data in the 90 decimal-digit precision, and the Taylor series in the different M (color online)

How about the effects of round-off errors on chaotic systems? It should be emphasized that double precision is widely used in numerical simulations, which brings round-off errors at every time step that increase exponentially for chaotic dynamic systems. In order to investigate the effects of round-off errors, we add a random data at each time step with zero mean and the standard deviation 10^{-16}, while the 80th-order Taylor expansion ($M = 80$) is still used and all data are expressed in 90 decimal-digits so as to guarantee the negligible truncation error in the considered interval of time $t \in [0, 10]$.

Figure 10 shows the comparison between the reliable statistics given by the CNS (using the 90 decimal-digit multiple precision for every datum) and the unreliable statistics given by double precision. Note that the round-off error has a great effect on the standard deviation of $A(t)$ from the very beginning. Within $0.0 \leqslant t \leqslant 3.5$, the difference in the standard deviation of $A(t)$ is so small that no obvious separation of mean is observed. However, at about $t = 3.5$ when the round-off error enlarges exponentially so that the standard deviation reach the level of the "true" chaotic solution, the mean of $A(t)$ becomes unreliable. Note that the mean of $A(t)$ given by the double precision becomes time-independent more early, at about $t = 3.5$, which is, however, wrong in physics, since the correct mean $A(t)$ of the chaotic system should become time-independent when $t > 7.0$, according to our reliable CNS result. As the system truly becomes unsteady when $t > 7.0$, the round-off error has no effect on statistics. Furthermore, Fig. 11 shows the scatter diagram of the PDF in the AD-plane of the chaotic dynamic system of three DOFs (see Eq. (14)) given by the two different numerical schemes. Note that the PDFs are almost the same at $t = 8.00$. However, the PDFs are obviously different at $t = 6.60$. All of these suggest that the round-off error might have a great effect on the unsteady statistics when the chaotic dynamic system is in a transition stage from an equilibrium state to a new one, while has a little effect on steady ones.

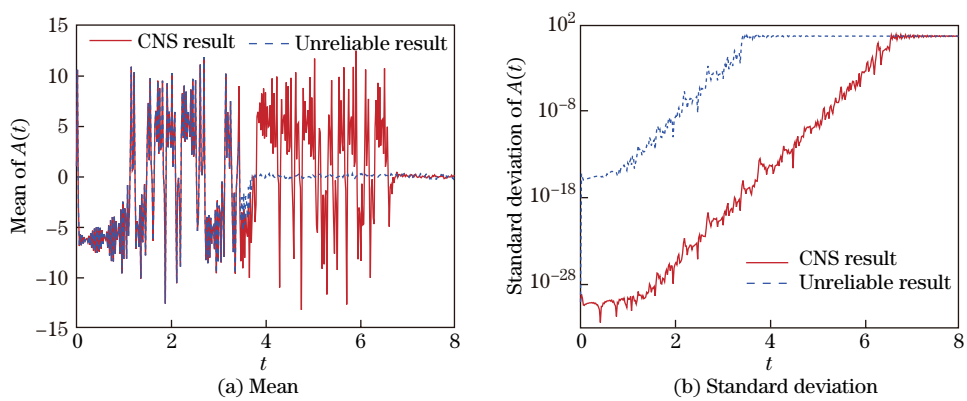
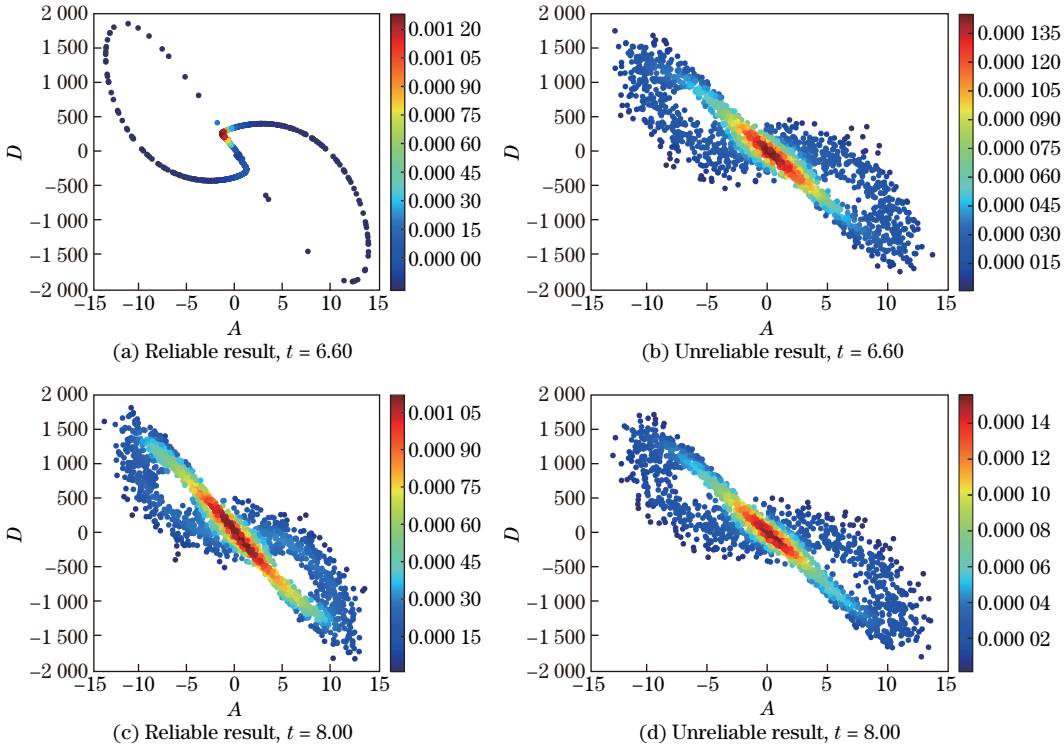


Fig. 10 Effects of the round-off error on the mean and the standard deviation of $A(t)$ of the chaotic system of three DOFs (see Eq. (14)) using the time step $h = 10^{-3}$ and the 80th-order Taylor expansion ($M = 80$) with the negligible truncation error, where solid lines in red denote the reliable CNS results given by the 90 decimal-digit multiple precision for every datum, while dashed lines in blue denote the unreliable results given by means of double precision (color online)

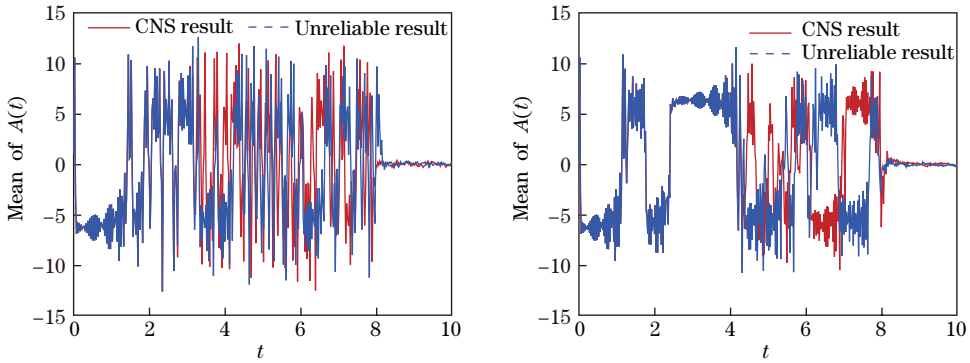
Similarly, we also investigate the effects of numerical noises on the computations of statistics of the chaotic dynamic systems of five and seven DOFs governed by Eqs. (15) and (16), respectively, and obtain the same conclusions qualitatively, as shown in Fig. 12. It suggests that the above-conclusions should have general meanings and would be qualitatively the same even for chaotic systems with large numbers of DOFs. This indicates that numerical noises might have a great effect even on the statistic properties if chaotic dynamic systems are time-dependent in statistics. The same conclusion has also been reported for chaotic three-body system in the literature[38]. Therefore, the conclusion has general meanings.



(a) Reliable result, t = 6.60 (b) Unreliable result, t = 6.60

(c) Reliable result, t = 8.00 (d) Unreliable result, t = 8.00

Fig. 11 Scatter diagrams of the PDF in the AD-plane of the chaotic system with three DOFs (see Eq. (14)) at different time, where the reliable results are given by the CNS using the 90 decimal-digit multiple precision for every datum, while the unreliable results are given by double precision (color online)



(a) Five DOFs (b) Seven DOFs

Fig. 12 Effects of the truncation error on the mean of $A(t)$ of the chaotic systems of five and seven DOFs obtained using the time step $h = 10^{-3}$ and the 90 decimal-digit multiple precision for every datum with the negligible round-off error, where solid lines in red denote the reliable results given by the CNS using the 80th-order Taylor series expansion for t, i.e., $M = 80$ with the negligible truncation error, while dashed lines in blue denote the unreliable results given by the 10th-order Taylor series expansion, i.e., $M = 10$ with the considerable truncation error (color online)

5 Concluding remarks and discussion

In this paper, we illustrate that the convergent (reliable) simulations of chaotic dynamic systems cannot be obtained by means of the traditional numerical algorithms in double pre-

cision, even including SIs, although they can conserve the total energy of the chaotic systems quite well. Therefore, the conservation of total energy itself cannot guarantee the reliability of trajectories, Fourier power spectra, and ACFs of chaotic dynamic systems. Such numerical phenomena lead to some intense arguments about the reliability of numerical simulations of chaos[5–6,8–9].

These arguments can be calmed down by means of the CNS. Based on an arbitrary order of Taylor series expansion and the data in arbitrary accuracy of multiple precision plus a convergence (reliability) check by means of comparison with an additional simulation with even small numerical noises, the CNS can provide us a new strategy to reduce the numerical noises to any required level so that the numerical noises are negligible in a long (but finite) interval of time, e.g., the CNS can give convergent reliable chaotic trajectories in a quite long interval of time for the chaotic Hénon-Heiles system and the famous three-body problems. It should be emphasized that, for the Hénon-Heiles system and the three-body problem, it is very important to give an accurate prediction of orbits. Therefore, the CNS can indeed bring us something completely new/different, although it is a kind of remixing of some known methods/technologies.

The SDIC of chaotic dynamic systems is well-known, which can be traced back to Poincaré[1] in 1880s and Lorenz[2] in 1960s who gave the SDIC a more popular name, i.e., the butterfly-effect. Even so, it is nowadays widely believed that the statistic properties of chaotic dynamic systems could be correctly obtained, even if convergent chaotic trajectories are impossible by means of traditional algorithms in double precisions. However, strictly speaking, this is only a kind of conjecture, i.e., an ideal wish. In this paper, we illustrate that even the statistic properties of chaotic systems cannot be correctly obtained by means of traditional numerical algorithms in double precision, as long as these statistics are time-dependent! The same conclusion has also been reported for chaotic three-body system in the literature[38]. Therefore, the conclusion has general meanings.

Note that the above-mentioned conclusions are based on some simple chaotic systems such as the Lorenz equation, the three-body problem, and the Hamiltonian Hénon-Heiles system. How about more complicated dynamic systems with an infinite number of dimensions, e.g., turbulent flows? Note that the famous Lorenz equation with three DOFs is a simplified model, which can be derived from the exact Navier-Stokes equation describing the two-dimensional RB convection flow[61]. Currently, Lin et al.[39] successfully applied the CNS to the two-dimensional RB convection flow for solving a dynamic system of 16 129 ($= 127 \times 127$) DOFs by means of the CNS. They provided a theoretical evidence that the exact Navier-Stokes equation has the SDIC and thus is chaotic. So, it is quite possible that numerical noises might have a great effect on the statistic properties of turbulent flows when these statistics are time-dependent. Thus, our CNS results strongly suggest that we had better to be careful on the results of statistically unsteady turbulent flows given by the DNS in double precision, although DNS results often agree well with experimental data when turbulent flows are in a statistical stationary state, as reported in Refs. [12] and [16].

Obviously, with negligible numerical noises in a long enough interval of time, the CNS could provide us a better and more reliable way to investigate chaotic dynamic systems. It should be emphasized that, by means of the CNS, we gave, for the first time, a convergent (reliable) chaotic trajectory of the Lorenz equation in a quite long interval $[0, 10\,000]$, which has been never obtained by any traditional algorithm in double precision[35]. Besides, the CNS provides, for the first time, a theoretical evidence[39] that the Navier-Stokes equations have the SDIC and are chaotic so that micro-level thermal fluctuation becomes the origin of the macroscopic randomness of RB turbulent flows, i.e., the randomness of turbulent flows is self-excited, out of nothing. In addition, hundreds of new periodic orbits of the three-body system with equal mass and zero momentum and over a thousand periodic orbits of planar three-body system with unequal mass have been found by means of the CNS for the first time[40,42]. Indeed, the CNS can bring us something completely new/different, although it is only a remixing of some

known methods/technologies that can be traced back even to Newton.

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